Función Transferencia

- $I_{R_1} = \dfrac{V_1 - V^-}{R_1}$, $I_{R_2} = \dfrac{V^- - V_2}{R_2}$, $V^- = V^+$

$$\Longrightarrow \frac{V_1 - V^-}{R_1} = \frac{V^- - V_2}{R_2}$$

$$\frac{V_1}{R_1} + \frac{V_2}{R_2} = \left(\frac{1}{R_1} + \frac{1}{R_2}\right) V^+$$

$$\Longrightarrow \boxed{V^+ = \frac{R_1 V_2 + R_2 V_1}{R_1 + R_2}}$$

- $I_{C_1} = \dfrac{V_1 - V^+}{1/sC}$, $I_{R_3} = \dfrac{V^+}{R_3}$

$$\Longrightarrow \frac{V_1 - V^+}{1/sC} = \frac{V^+}{R_3}$$

$$sCV_1 = \left(sC + \frac{1}{R_3}\right) V^+$$

$$sCR_3V_1 = (sCR_3 + 1) \cdot \frac{R_1V_2 + R_2V_1}{R_1 + R_2}$$

$$(R_1 + R_2)sCR_3V_1 - (sCR_3 + 1)R_2V_1 = (sCR_3 + 1)R_1V_2$$

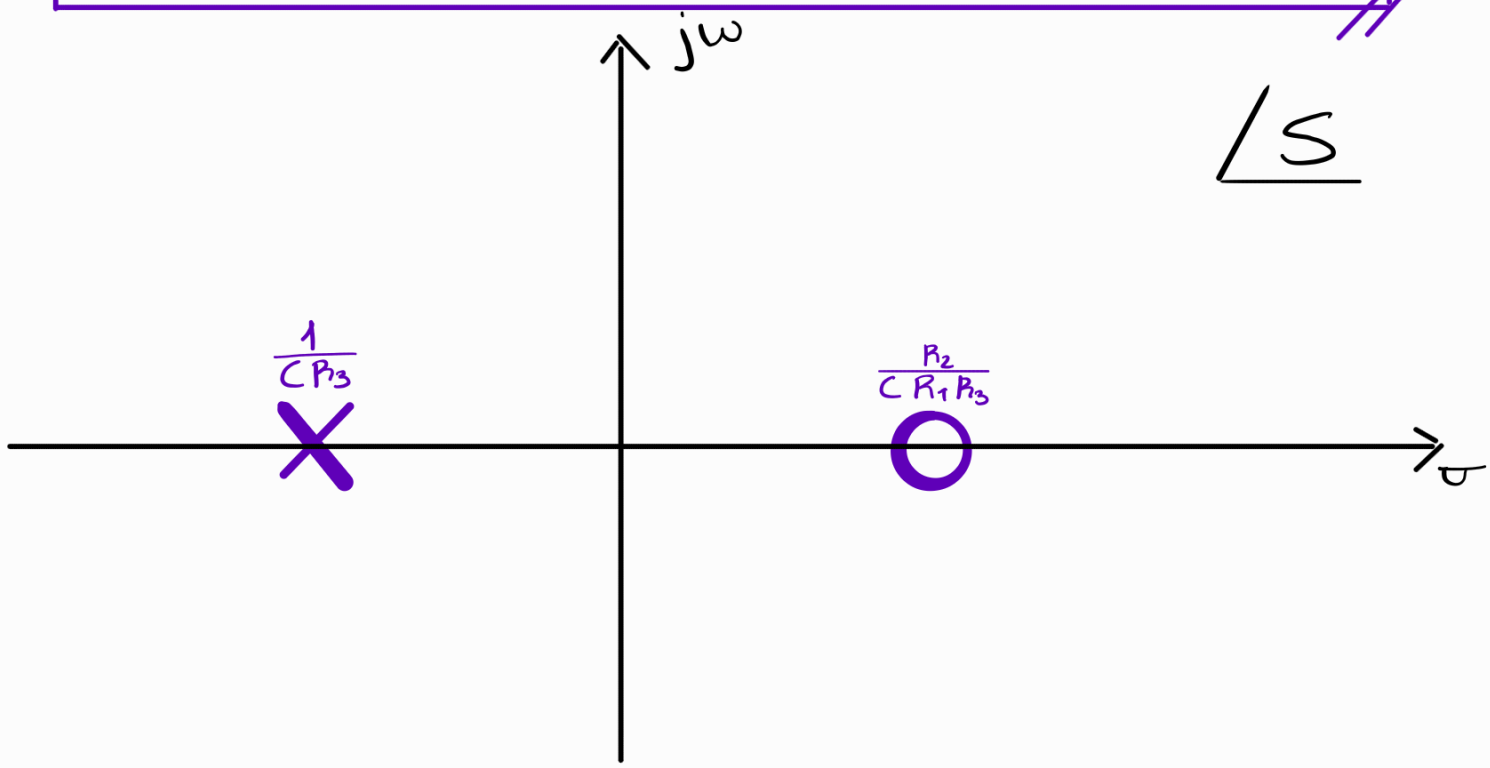
$$\left[sCR_1R_3 + \cancel{sCR_2R_3} - \cancel{sCR_2R_3} - R_2\right]V_1 = CR_1R_3\left(s + \frac{1}{CR_3}\right)V_{2_3}$$

$$T(s) = \frac{V_2}{V_1} = \frac{sCR_1R_3 - R_2}{CR_1R_3\left(s + \frac{1}{CR_3}\right)}$$

$$\boxed{\frac{V_2}{V_1} = \frac{s - \frac{R_2}{CR_1R_3}}{s + \frac{1}{CR_3}}}$$

$$\left.\frac{V_2}{V_1}\right|_{s=j\omega} = \frac{\frac{-R_2}{CR_1R_3} + j\omega}{\frac{1}{CR_3} + j\omega}$$

$$\left.\frac{V_2}{V_1}\right|_{s=j\omega} = \frac{\sqrt{\left(\frac{R_2}{CR_1R_3}\right)^2 + \omega^2}}{\sqrt{\left(\frac{1}{CR_3}\right)^2 + \omega^2}} \cdot e^{j\left(\measuredangle\left(\frac{V_2}{V_1}\right)\Big|_{s=j\omega}\right)}$$

$$\measuredangle\left(\frac{V_2}{V_1}\right)\Big|_{s=j\omega} = \operatorname{arctg}\left(-\frac{CR_1R_3}{R_2}\cdot\omega\right) - \operatorname{arctg}\left(CR_3\cdot\omega\right)$$

$$\boxed{\begin{aligned} \left|\frac{V_2}{V_1}\right|\Big|_{s=j\omega} &= \frac{\sqrt{\left(\frac{R_2}{CR_1R_3}\right)^2 + \omega^2}}{\sqrt{\left(\frac{1}{CR_3}\right)^2 + \omega^2}} \\ \measuredangle\frac{V_2}{V_1}\Big|_{s=j\omega} &= \operatorname{arctg}\left(-\frac{CR_1R_3}{R_2}\cdot\omega\right) - \operatorname{arctg}\left(CR_3\cdot\omega\right) \end{aligned}}$$